

Predictive Workforce Analytics: Graduate Pay & Placement Intelligence

A big data approach to compensation benchmarking using Singapore’s Graduate Employment Survey

Course Big Data Analytics	Assessment Group Project (20 marks)
Programme MBA (Business Analytics), Trimester IV	Dataset NUS Graduate Employment Survey 2024, MOE Singapore
Institution (Hidden — Report by Abhishek)	Platform Apache Spark (PySpark MLlib) on Databricks

1. Problem Statement

Organisations hiring fresh graduates set compensation and target recruitment largely on instinct and inherited spreadsheets. A single company-wide “graduate salary band” is commonly applied across degree programmes whose actual market rates differ by a factor of two, and applications are screened using GPA cut-offs that have never been validated against pay or placement outcomes. The consequence is systematic mispricing — overpaying some hires while losing others to rejected offers — compounded by recruitment concentrated in a small number of faculties where competition is most intense and availability lowest.

Singapore’s Ministry of Education publishes detailed graduate employment and salary outcomes annually, yet the data is rarely used beyond reading its headline figures. It is released as aggregated summary statistics rather than individual records, which places it outside the reach of conventional predictive modelling and leaves its analytical value largely unrealised.

This project therefore asks whether a publicly available, aggregate-only employment survey can be transformed into an operational predictive tool capable of answering two questions that hiring teams face every recruitment cycle: **what a given graduate should be paid**, and **which graduate pools remain available in the market**. Addressing this requires reconstructing an individual-level dataset from published distributional statistics, engineering features at scale, and building and critically evaluating machine learning models — the full big data analytics workflow applied to a genuine business decision problem.

2. Research Questions

RQ1	How much of the variance in graduate salary is explained by degree choice, relative to individual attributes such as GPA, internship experience and honours classification?
RQ2	Can gross monthly salary be predicted to a tolerance narrow enough to be useful for setting graduate offer bands?
RQ3	Can full-time permanent placement be predicted from attributes observable at the point of graduation, and does reported model accuracy withstand scrutiny under class imbalance?
RQ4	Where does the published survey mislead its readers, and which variables would most improve predictive power if collected?

3. Mapping to Course Learning Objectives

Each course learning objective is addressed by a specific, assessable component of the project. Programme learning objectives are shown in parentheses.

CLO	Objective	How this project addresses it	Evidence produced
CLO 1 (PLO 2a)	Understand the concept of big data and its use in a business environment	Frames graduate compensation as a business problem before any technology is introduced, and establishes why aggregate national survey data cannot answer it without reconstruction and distributed processing.	Problem framing; market-structure analysis showing two non-overlapping salary distributions
CLO 2 (PLO 3b)	Create and develop critical skills in handling and managing big data	Reconstructs 4,340 individual graduate profiles from published quartiles using fitted log-normal and Bernoulli sampling; ingests and manages the dataset through a reproducible Spark pipeline.	Data generation pipeline; Spark DataFrame ingestion; feature engineering pipeline (StringIndexer → OneHotEncoder → VectorAssembler → StandardScaler)
CLO 3 (PLO 3c)	Develop and apply an integrative approach to decision making and problem solving	Conducts cause-and-effect analysis linking degree structure to pay and placement outcomes, then converts findings into costed recommendations for employers, universities and policymakers.	Feature-importance analysis; mean-versus-median distortion finding; five stakeholder recommendations
CLO 4 (PLO 4c)	Evaluate and develop technological orientation, digital fluency and programming skills	Implements and comparatively evaluates eight models across regression and classification in PySpark MLlib, with held-out testing, residual analysis and diagnosis of a class-imbalance artefact.	Four regressors and four classifiers; RMSE / MAE / R^2 and AUC / accuracy / F1 comparison; confusion matrix and ROC diagnostics

4. Scope and Approach

Data	NUS Graduate Employment Survey 2024 (MOE Singapore) — 7,382 respondents, 28 degree programmes, surveyed six months after graduation.
Reconstruction	4,340 individual profiles generated by sampling from log-normal distributions fitted to each programme's published 25th, 50th and 75th percentiles; placement drawn from published employment rates.
Modelling	Regression on gross monthly salary and binary classification on full-time permanent placement; 80/20 held-out split, fixed seed, four algorithms per task.
Evaluation	RMSE, MAE and R^2 for regression; AUC, accuracy, F1, confusion matrix and ROC for classification, benchmarked against a no-skill baseline.

5. Anticipated Outcomes

- A quantified decomposition of graduate salary variance identifying which attributes genuinely predict pay.
- A salary prediction model with an error tolerance stated in Singapore-dollar terms usable for offer banding.
- An honest assessment of placement predictability, including whether headline accuracy survives a no-skill baseline comparison.
- A set of costed, stakeholder-specific recommendations, and identification of the highest-value variable absent from the national survey.

Limitation acknowledged at the outset: individual records are reconstructed from published aggregate statistics because privacy legislation restricts release of the underlying microdata. Aggregate statistics are authentic; individual observations are simulated to preserve them.